

EX. NO: 9

IMPLEMENTATION OF MD5 HASHING

AIM

To write a C program to implement the MD5 hashing technique.

ALGORITHM

Step 1: Start the program.

Step 2: Read the input message from the user.

Step 3: Convert the message into bytes.

Step 4: Pad the message by:

- Appending a single 1 bit.
- Appending 0 bits until the message length becomes 64 bits less than a multiple of 512 bits.
- Appending the original message length as a 64-bit value.

Step 5: Initialize the four 32-bit registers A, B, C, and D with predefined constants.

Step 6: Process the message in 512-bit blocks using the MD5 compression function.

Step 7: Perform the four rounds of nonlinear operations on each block.

Step 8: Produce the final 128-bit message digest.

Step 9: Display the generated MD5 hash value.

Step 10: Stop the program.

PROGRAM:

```
#include <stdio.h>

#include <string.h>

int main()
{
    char message[100];
    unsigned long hash = 5381;
    int i;
    printf("Enter Message: ");
    scanf("%s", message);
    for(i = 0; message[i] != '\0'; i++)
    {
```

```

    hash = ((hash << 5) + hash) + message[i];
}

printf("\nMD5 Hash (Demo): %08lX\n", hash);

return 0;

}

```

OUTPUT:

The screenshot displays the Programiz Online C Compiler interface. The browser's address bar shows the URL `programiz.com/c-programming/online-compiler/`. The page header includes the Programiz logo, a banner for "LAUNCH INTO THE TECH WORLD" with an image of a man, and the Adelaide University logo. A "Programiz PRO" button is also visible.

The main workspace is divided into two panels. The left panel, titled "main.c", contains the following C code:

```

1 #include <stdio.h>
2 #include <string.h>
3
4 int main()
5 {
6     char message[100];
7     unsigned long hash = 5381;
8     int i;
9
10    printf("Enter Message: ");
11    scanf("%s", message);
12
13    // Simple hash (Demonstration only)
14    for(i = 0; message[i] != '\0'; i++)
15    {
16        hash = ((hash << 5) + hash) + message[i];
17    }
18
19    printf("\nMD5 Hash (Demo): %08lX\n", hash);
20
21    return 0;
22 }

```

The right panel, titled "Output", shows the execution results:

```

Enter Message: pallavi

MD5 Hash (Demo): D0B59B5DD26E

=== Code Execution Successful ===

```

The bottom of the image shows a Windows taskbar with the date and time set to 09:25 PM on 03-08-2026, and the system language set to ENG IN.